Exercise Sheet

Lecture 7

Inner Product Spaces

The inner product is taken in the physics convention: conjugate-linear in the first slot, linear in the second, so $\langle \phi \mid \psi \rangle = \langle \phi, \psi \rangle$ is linear in the ket. The core spaces are finite-dimensional. In quantum questions, normalized kets represent pure-state rays and measurements are ideal projective ones. Tags mark the flavour; a ⋆ marks a harder one.

Core tutorial route

Work **7.5, 7.1, 7.17, 7.21, 7.23, 7.2, and 7.11** in that order (about 80–100 minutes before discussion). This seven-problem route covers axiom diagnosis, finite complex norm calculation, orthogonal projection, a Cauchy–Schwarz proof, the distinct triangle-equality condition, complex polarization in the physics convention, and finite quantum transfer.

Additional practice: 7.4, 7.6, 7.8, 7.14–7.16, 7.18–7.20, 7.22, 7.24–7.25. **Bridge:** 7.3, 7.7, 7.12–7.13 (integral models, a rank-one projector, and finite ONB expansion). **Extension:** 7.9–7.10 (repeated measurement and expectation drills). Bridge and extension tasks do not count toward core progress.

Inner products and norms

Exercise 7.1 (computational) For $u = (1, -i, 2)$ and $v = (i, 1, 1 - i)$ in $\mathbb{C}^3$ with the standard inner product, compute $\langle u, v\rangle$, $\langle v, u\rangle$, $\|u\|$, and $\|v\|$.

Answer. $\langle u, v\rangle = 2$, $\langle v, u\rangle = 2$, $\|u\| = \sqrt{6}$, $\|v\| = 2$.

Exercise 7.2 (proof) Use the complex polarization identity in the physics convention to recover $\langle u, v\rangle$ from norms when $u = (1, i)$ and $v = (2, 1)$ in $\mathbb{C}^2$. Show all four terms $\|v + \alpha u\|^2$ for $\alpha \in \{1, -1, i, -i\}$ and check your result directly.

Answer. $\langle u, v\rangle = 2 - i$; the four squared norms are $11, 3, 5, 9$ in the order $1, -1, i, -i$.

Exercise 7.3 (computational) On $C[0, 1]$ with $\langle f, g\rangle = \int_0^1 \overline{f} g \, dt$, compute $\langle 1, t\rangle$, $\|1\|$, $\|t\|$, and verify the Cauchy–Schwarz inequality for these two functions.

Answer. $\langle 1, t\rangle = \frac{1}{2}$, $\|1\| = 1$, $\|t\| = \frac{1}{\sqrt{3}}$; Cauchy–Schwarz $\frac{1}{2} \le \frac{1}{\sqrt{3}}$ holds.

Exercise 7.4 (computational) For $u = (1, 2, 2)$ and $v = (2, 2, 1)$ in $\mathbb{R}^3$ with the standard inner product, compute $\langle u, v\rangle$, $\|u\|$, $\|v\|$, and the cosine of the angle between them.

Answer. $\langle u, v\rangle = 8$, $\|u\| = \|v\| = 3$, $\cos\theta = \frac{8}{9}$.

Exercise 7.5 (proof) On $\mathbb{C}^2$ consider the proposed form $B(u, v) = \overline{u_1} v_1 - \overline{u_2} v_2$. Decide whether B is an inner product in the physics convention. Justify conjugate symmetry, linearity in the second slot, and positive definiteness separately, identifying every satisfied or failed axiom.

Answer. B is conjugate-symmetric and linear in the second slot, but is not positive definite; $B((0, 1), (0, 1)) = -1$.

Exercise 7.6 (computational) On $\mathcal{M}_2(\mathbb{R})$ with the Frobenius inner product $\langle A, B\rangle = \mathrm{tr}(A^\dagger B)$, take $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Compute $\langle A, B\rangle$, $\|A\|$, and $\|B\|$.

Answer. $\langle A, B\rangle = 4$, $\|A\| = 3$, $\|B\| = \sqrt{6}$.

Exercise 7.7 (computational) On $C[0,1]$ with $\langle f, g\rangle = \int_0^1 \overline{f} g\, dt$, let $f(t) = 1+t$ and $g(t) = 1-t$. Compute $\langle f, g\rangle$, $\|f\|$, and $\|g\|$.
Answer. $\langle f, g\rangle = \frac{2}{3}$, $\|f\| = \sqrt{7/3}$, $\|g\| = \frac{1}{\sqrt{3}}$.

Exercise 7.8 (proof) In a *real* inner product space, prove that if $\|u\| = \|v\|$ then $u + v$ is orthogonal to $u - v$ (the diagonals of a rhombus are perpendicular).

Dirac notation

Exercise 7.9 (computational) In an orthonormal basis $|0\rangle, |1\rangle$, let $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$. Find the probability of each outcome in a measurement in the $\{|0\rangle, |1\rangle\}$ basis, and the expectation value of $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$.
Answer. Probabilities $\frac{9}{25}$ and $\frac{16}{25}$; $\langle \sigma_z\rangle = -\frac{7}{25}$.

Exercise 7.10 (computational) In an orthonormal basis $|0\rangle, |1\rangle$, let $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$. Find the probability of each outcome in the $\{|0\rangle, |1\rangle\}$ basis and the expectation value of $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$.
Answer. Probabilities $\frac{1}{4}$ and $\frac{3}{4}$; $\langle \sigma_z\rangle = -\frac{1}{2}$.

Exercise 7.11 (computational) In an orthonormal basis $|0\rangle, |1\rangle$, let $|\psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + i|1\rangle\big)$ and $|+\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)$. Find the probabilities of $|0\rangle, |1\rangle$ in the $\{|0\rangle, |1\rangle\}$ basis, the amplitude $\langle + \mid \psi\rangle$, and the probability $|\langle + \mid \psi\rangle|^2$.
Answer. Probabilities $\frac{1}{2}$ and $\frac{1}{2}$; $\langle + \mid \psi\rangle = \frac{1+i}{2}$, $|\langle + \mid \psi\rangle|^2 = \frac{1}{2}$.

Exercise 7.12 (computational) With $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$, write the matrix of the outer product $|\psi\rangle\langle\psi|$ in this basis for $|\psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)$.
Answer. $|\psi\rangle\langle\psi| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Exercise 7.13 (proof) Let $\{|n\rangle\}$ be an orthonormal basis and $|\psi\rangle = \sum_n c_n |n\rangle$. Prove that $\langle m \mid \psi\rangle = c_m$ and that $\|\psi\|^2 = \sum_n |c_n|^2$.

Exercise 7.14 (proof) Show that the bra associated to $a|\phi\rangle + b|\chi\rangle$ is $\overline{a}\langle\phi| + \overline{b}\langle\chi|$.

Orthogonality and Cauchy–Schwarz

Exercise 7.15 (computational) Find all $a \in \mathbb{C}$ for which $(1, a)$ is orthogonal to $(1, i)$ in $\mathbb{C}^2$.
Answer. $a = -i$.

Exercise 7.16 (computational) Find the value of $a \in \mathbb{R}$ for which $(1, a, 2)$ is orthogonal to $(3, -1, 1)$ in $\mathbb{R}^3$ with the standard inner product.
Answer. $a = 5$.

Exercise 7.17 (computational) In $\mathbb{R}^2$ decompose $u = (4, 2)$ as a multiple of $v = (1, 1)$ plus a vector orthogonal to v, using $\dfrac{\langle v, u\rangle}{\|v\|^2}\, v$ for the projection.

Answer. $u = (3,3) + (1,-1)$, with $(3,3) = 3v$ along v and $(1,-1) \perp v$.

Exercise 7.18 (computational) For $u = (1,1)$ and $v = (1,i)$ in $\mathbb{C}^2$, compute $|\langle u,v\rangle|$, $\|u\|$, and $\|v\|$, and verify the Cauchy–Schwarz inequality.

Answer. $|\langle u,v\rangle| = \sqrt{2}$, $\|u\| = \|v\| = \sqrt{2}$; Cauchy–Schwarz $\sqrt{2} \le 2$ holds.

Exercise 7.19 (computational) In $\mathbb{R}^4$ find the cosine of the angle between $u = (1,1,1,1)$ and $v = (1,1,1,-1)$, and hence the angle.

Answer. $\cos\theta = \frac{1}{2}$, $\theta = 60°$.

Exercise 7.20 (computational) Compute the Gram matrix $G_{jk} = \langle v_j, v_k\rangle$ of $v_1 = (1,1,0)$ and $v_2 = (0,1,1)$ in $\mathbb{R}^3$, and use it to decide whether they are linearly independent.

Answer. $G = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $\det G = 3 \neq 0$, so independent.

Exercise 7.21 (proof) Prove Cauchy–Schwarz by treating $v = 0$ separately and, for $v \neq 0$, decomposing $u = \frac{\langle v,u\rangle}{\|v\|^2} v + w$ with $w \perp v$. State exactly when equality holds.

Exercise 7.22 (proof) Use the Cauchy–Schwarz inequality to prove that for $a_1, \dots, a_n \in \mathbb{C}$,

$$\left|\sum_{k=1}^{n} a_k\right|^2 \le n \sum_{k=1}^{n} |a_k|^2.$$

Exercise 7.23 (⋆) Prove that equality holds in the triangle inequality $\|u+v\| = \|u\| + \|v\|$ if and only if one of u, v is a nonnegative real multiple of the other.

Hint. Square the proposed equality and compare with the expansion of $\|u+v\|^2$; this lands you in the Cauchy–Schwarz equality case, after which the phase of the scalar is forced.

Exercise 7.24 (⋆) Let $v_1, \dots, v_m$ be nonzero pairwise-orthogonal vectors. Prove they are linearly independent.

Hint. Pair a vanishing linear combination with each v_j; orthogonality kills every term but one.

Exercise 7.25 (proof) For a subspace $U \subseteq V$ of an inner product space, prove that $U \cap U^\perp = \{0\}$.